The Influence of Strength Level on Handball-Specific Fitness Elements Between Stronger and Weaker Professional Players

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Abstract

Kyriacou-Rossi, A, Hadjicharalambous, M, and Zaras, N. The influence of strength level on handball-specific fitness elements between stronger and weaker professional players. *J Strength Cond Res* 38(5): 966–975, 2024—The aim of the study was to investigate the influence of strength level between stronger and weaker handball players on handball specific fitness elements and to explore the correlations between strength and sport-specific fitness elements. Twenty-one professional male handball-players (age: 25.9 ± 6.9 years; mass: 87.9 ± 13.9 kg; height: 1.81 ± 0.08 m), participated in the study. Players were divided into the stronger (SG) and weaker group (WG) according to their relative to body mass 1 repetition maximum strength (1RM). Measurements included body composition, countermovement jump (CMJ), isometric leg extension peak torque (IPT) and rate of torque development (RTD), 5-step long-jump, 0–20 m linear sprint, T-half test, throwing velocity, and 1RM in bench press and squat. No significant difference was found for body composition between SG and WG ($p > 0.05$). However, SG had significantly higher CMJ height (21.5%, $p = 0.002$), IPT (22.4%, $p = 0.008$), RTD relative to body mass ($p < 0.05$), 5-step long jump (10.9%, $p = 0.005$), lower 0–20 linear sprint (−6.3%, $p = 0.012$), lower T-half test time trial (−7.3%, $p = 0.001$), and higher throwing velocity compared with WG ($p < 0.05$). When all players included in one group, large to very large correlations were found between 1RM strength and IPT with fat-free mass ($r = 0.518$–0.774) and throwing velocity ($r = 0.472$–0.819). Very large correlations were found between RTD with fat-free mass ($r = 0.760$) and throwing velocity ($r = 0.780$–0.835). Stronger players have greater performance in all handball-fitness attributes compared with their weaker counterparts. The significant correlations between handball-specific fitness elements with strength and RTD suggest that strength training is essential for handball players as it may link to higher on court performance.

Key Words: team-sport, fat-free mass, rate of torque development, throwing velocity

Introduction

Handball is a multicomponent sport that requires high-level physical abilities and well-developed sport-specific skills for attaining success. Therefore, performance in such a demanding sport may be determined by numerous factors including the individual fitness level and technical skills as well as the team tactical strategy (37,43). Along with the specific, highly demanding technical and tactical skills required during a handball game, players perform a variety of explosive movements that require vigorous muscular effort (4,16,37). Such movements comprise of forward and backward shuttles, rapid changes of directions, jumping, turning, high-speed ball shooting and blocking, as well as hard physical contacts either during the offensive or the defensive phase of the game (12,13,19). Consequently, handball players devote a large proportion of their daily training to enhance throwing, jumping, and sprinting abilities as well as to resistance training to increase muscular strength and power in an attempt to improve performance during the game.

Muscle strength is a determinant factor for performance in several individual (40) and team sports (5) and, therefore, in handball (15,16,24,37,43). In addition, several studies have shown that elite or higher-level handball players have greater body mass, possess greater fat-free mass, are stronger, faster, more powerful, and have greater throwing velocity compared with nonelite or to lower-level players (11,29). However, apart from the competitive level of players, several studies have shown that there is a strong difference between stronger and weaker athletes on both physiological and fitness parameters. Specifically, stronger athletes possess greater muscle mass and muscle thickness compared with their weaker counterparts (21,32) producing, therefore, a greater amount of maximum force and power as well as a higher rate of force development (RFD) and isometric peak force (IPF) compared with weaker athletes (21,34,35,42). According to the author's knowledge, the influence of strength level on handball-specific fitness elements remains largely unexplored, meaning that whether stronger handball players may have a greater advantage in handball-specific elements compared with their weaker counterparts needs further investigation. The answer of this question will help coaches to better understand the role of player's strength level in handball-specific fitness elements. Thus, coaches will be able to design more specific and yet more effective resistance training programs according to the player's strength level.

Long-term resistance training programs may induce significant increases in strength in handball players (12,17,24). These increases are accompanied with positive changes in throwing velocity; however, the results are ambiguous for sprint times and vertical jump performance (10,15,17), mainly because of the nature of the different resistance training programs applied in the

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studies. Although running and jumping are essential handball fitness skills, throwing velocity combined with throwing accuracy is considered being the most important technical skill in handball as it is a major factor for all actions of the players in the court (39). Results from studies have shown that throwing velocity was significantly correlated with 1 repetition maximum (1RM) in bench press and in lower-body force production (3,12,23). Considering that strength positively contributes to throwing velocity and overall handball performance (15,37), a more intriguing question would be whether RFD and IPF may correlate with handball-specific skills. Rate of force development is believed to be a key factor influencing successful performance in power sports (22,41) and can be assessed by the force-time curve in short time periods ranging from 0 to 250 milliseconds during both dynamic and isometric muscle actions (6,42). The rate of force development may provide useful insights regarding the neuromuscular function and readiness of athletes before a competition (39,40) because it is strongly depend on motor unit recruitment and discharge rate as well as on muscle fiber type composition and myofibrillar mechanisms (6,22). High RFD is also related to high impulse which represents a more comprehensive image of the athletes' velocity capacity in specific time windows (33). Several studies have linked RFD with athletic performance (26,40–42), although the relationship between RFD and handball-specific fitness elements is largely unexplored.

The purpose, therefore, of this study was to investigate the influence of strength level on handball-specific fitness elements between stronger and weaker players and to examine the relationship between strength and RFD with handball-specific fitness elements. The hypothesis of the study was that stronger players will have a higher performance on handball-specific fitness elements compare with their weaker counterparts, whereas strength and RFD will significantly be correlated with handball-specific fitness elements.

Methods

Experimental Approach to the Problem

This is a cross-sectional study dealing with the observation of handball players 2 weeks before the initiation of the official competition period (end of the preseason training period). Experimental procedures occurred during a 4-day period. During the first day, players visited the laboratory for the evaluation of body composition and a familiarization session in countermovement jump (CMJ), and isometric leg extension rate of torque development (RTD) and peak torque (IPT). During the second day, players visited the laboratory for the official measurements of CMJ, IPT, and RTD. The next day, field testing measurements were performed including the 5-step long-jump, 0–20 m linear sprint, and T-half test change of direction time trial. During the last day of measurements, standing throwing velocity from 7-m penalty line and 9-m jumping shot were measured and 1RM strength in bench press and squat were evaluated. After the measurements, players were separated either into a strong (SG: n = 10) or a weak (WG: n = 11) group according to the median score of total 1RM strength relative to body mass in bench press and squat (SG: 2.91 ± 0.38 $kg \cdot kg^{-1}$ vs. WG: 2.22 ± 0.37 $kg \cdot kg^{-1}$, p = 0.001). Moreover, the 4 goalkeepers of the team were randomly divided equally into the 2 groups of strong and weak. Further analysis between groups showed that SG trained approximately for 10.8 ± 0.6 hours per week and WG trained approximately for 10.3 ± 1.3 hours per week, with no significant difference between groups (p = 0.263).

Subjects

Twenty-one male professional handball players (age: 25.9 ± 6.9 years, age ranged from 17 to 35 years of age; height: 1.81 ± 0.08 m; body mass: 87.9 ± 13.9 kg), with a 12.3 ± 3.2 years training experience, members of one of the leading teams in the National championship participated in the study. The study occurred 2 weeks prior the initiation of the competitive season of 2022–2023. All players followed a 7-week training phase aiming to increase strength/power and specific fitness skills (running, jumping, and throwing), before the initiation of the study. Players identified as healthy and free from any cardiovascular, orthopedic, and neuromuscular injuries. Players were informed of the benefits and risks of the investigation before signing an institutionally approved informed consent document to participate in the study, whereas for players under the 18th year of age, a parental or guardian signed consent was obtained. All procedures were in accordance to the 1975 Declaration of Helsinki as revised in 2000 and were approved by the National Bioethics Committee of Cyprus (EEBK/EΠ/2022/46).

Procedures

Anthropometric Characteristics and Body Composition. During the first day, players visited the laboratory for body mass, body height, and body composition analysis by the bioelectrical impedance scale. Body composition analysis (fat-free mass and percentage body fat) was performed during morning hours while players were instructed to fast for approximately 8 hours and abstained from any strenuous exercise during the previous day (36). In brief, body height of players was initially measured with a measuring tape which was positioned on the laboratory wall. Then, players stepped on the bioelectrical impedance scale (Tanita MC-780MA, Tokyo, Japan), only with their competition clothes without socks and shoes and stood still for approximately 45 seconds. After the body composition analysis, a familiarization session in CMJ, IPT, and RTD was performed.

Countermovement Jump, Isometric Peak Torque, and Rate of Torque Development. During the second day of laboratory measurements, the evaluations of CMJ, IPT, and RTD were performed. The evaluation of CMJ was performed first (OptoJump Next, Microgate, Bolzano, Italy). In brief, after a 5-minute warm-up on a stationary bike and a series of lower-body and upper-body dynamic stretching exercises, each player performed 3 warm-up CMJs with submaximal intensity. Subsequently, 4 maximum CMJs were performed with hands akimbo. During CMJs, players had the instruction to jump as high as possible, whereas 1 minute rest was given between jumps. After each jumping trial, researchers informed players about the jump height to motivate them for a greater effort. The best CMJ height was used for the statistical analysis. Ten minutes after the CMJs, the isometric leg extension torque was evaluated. Players seated on the isokinetic dynamometer chair (Humac Norm, CSMI, Stoughton, MA) and straps were used to ensure a stable body and exercising leg position. The exercising leg was determined with the Waterloo footedness questionnaire (7). Knee angle was set at 60° flexion (0°= full extension) as previously described (30). For the isometric evaluation, a warm-up procedure consisting of 3 submaximal efforts was performed, whereas players were instructed to progressively apply their force. Then, 3 maximal efforts were allowed with 2 minutes rest between attempts. In all attempts, players were instructed to apply their maximum force as fast as possible and to sustain it for 3 seconds. All data collected from the leg extension isometric measurements were recorded and analyzed to calculate the IPT and the RTD from the torque-time curve. IPT was calculated as the greater force generated from the torque-time curve, whereas RTD was calculated as the mean tangential slope of the torque time curve in specific time windows of 0–20, 0–60, 0–80, 0–100, 0–120, 0–150, 0–200, and 0–250 milliseconds. The best performance from the torque-time curve was used for the statistical analysis.

Five-Step Long-Jump Test. During the third day of measurements, players reported to the team training facilities for the evaluation of 5-step long-jump test, linear sprint, and T-half test. The 5-step long jump involves a series of 5 forward jumps with alternate left and right foot steps (2). Players performed 10 minutes running warm-up followed by whole-body dynamic stretching exercises. Then, 2 submaximal trials were given, and 3 maximal efforts with 2 minutes rest between them were performed. Players had the instruction to cover the greatest horizontal distance possible. Measurement was performed with a measuring tape to the nearest 0.1 cm, and the distance of the jumps was measured from the take-off point to the mark where the heels landed at the end of the fifth step. The longest 5-step jump was used for statistics.

Linear Sprint and T-Half Test. Ten minutes after the 5-step long-jump test, 0–20 m linear sprint was performed. In brief, after 2 submaximal sprints of 20 m, players performed 2 maximal sprints with a 5-minute rest between them. All sprint attempts were recorded with photocell gates (Witty, Microgate, Bolzano, Italy), which were placed approximately 1 m above the ground and 3 m apart. Each player started the sprint at 0.4 m behind the first gate on their own initiative. Of the 2 time trials, the best was used for statistics. Ten minutes after the linear sprint test, players performed the T-half test (17,31). Similar to the linear sprint test, players were given 2 submaximal and 2 maximal attempts with a 3-minute rest time in between. One photocell gate (Witty, Microgate, Bolzano, Italy) was set at the starting line approximately 1 m above the ground and 3 m apart, whereas players initiated their effort 0.4 m behind the starting line. For players who failed to maintain a facing forward position, crossed their feet, or failed to touch a cone during their attempt, then this attempt immediately stopped and was given an extra attempt. The best time trial performance was used for statistics. Moreover, as the athletes covered a 20 m distance both linear and by changing directions, the change of direction deficit (CoDd) was calculated, by subtracting from the time of T-half test to that of 20 m linear sprint (28) which is a more isolate measurement of changing direction (20).

Throwing Velocity. During the fourth day of measurements, the ball throwing velocity was evaluated. After 15 minutes of standardize warm-up including running, dynamic stretching (18), and several ball exercises (passes and shots), players performed maximum velocity throws from a standing position from the 7 m penalty line and jumping shot from the 9 m line. Throwing velocity was measured with a radar gun (Bushnell Velocity Sports Radar Gun, KS) measuring to the nearest 1.0 mph, whereas all throws were performed with a standard game ball (mass 480 g and circumference 58 cm), as previously described (38,39). The radar gun was positioned behind the goal, whereas all players were instructed to aim the ball between a target marked in front of the radar gun. Three to 4 warm-up throws were given and then 5 maximum throws with 2 minutes rest. Players performed first the standing throw from the penalty line (7 m) and then the jumping shot (9 m line). The best throwing velocity was used for the statistics.

1RM Strength. Fifteen minutes after the throwing velocity test, players performed the 1RM strength test in bench press and squat. First, the 1RM in bench press was evaluated, and 15 minutes later, the 1-RM in squat followed. The same method of progressively increasing loads was used for both exercises and included 2 sets with unloaded barbell for as many repetitions the players wanted, 1 set of 10 repetitions at approximately the 50% of the predicted 1RM, 1 set of 6 repetitions at the 70% of the predicted 1RM, and 1 set of 2–3 repetitions at the 90% of the predicted 1RM. Thereafter, players had 2–3 maximum attempts for the determination of 1RM with 3 minutes rest between sets (42). During all measurements, at least 2 of the researchers were present to evaluate the technique and encourage players to lift heavier loads. Players were instructed to lift the loads with maximum velocity during the concentric phase of movement, regardless of the actual movement velocity (1). For the bench press, a full range of motion was used, with the barbell been lowered until the chest and back up. For the squat, an elastic antenna was placed on an adjusted stool by the side of the player so that the thighs were parallel to the ground. When the pelvis touched, the antenna player began the upward movement.

Statistical Analyses

All variables are presented as means and *SD*s. Normality of data was assessed with the Shapiro-Wilk test, and no violations in normality were observed. A post hoc sample power analysis showed a power of 0.779 for the 21 players. Independent samples *T* test analysis was used to compare the differences between the SG and WG. Because of the small sample size, Hedges *g* effect size was calculated, with the following criteria used to infer the magnitude of the difference: <0.2 (trivial), 0.2–0.5 (small), 0.5–0.8 (moderate), and >0.8 (large) (8). Correlations between variables were examined with Pearson's correlation coefficient. Hopkins scales were used to investigate the magnitude of effect for the correlations: trivial <0.10, small <0.10–0.29, moderate ≤0.30–0.49, large ≤0.50–0.69, very large ≤0.70–0.89, and nearly perfect ≥0.9 (14). Standard multiple regression analysis was performed. Because of the small sample size ($n = 21$), adjusted R^2 was used for the interpretation of the multiple regression analysis results. The reliability of all measurements was determined with a 2-way random effect intraclass correlation coefficient, confidence intervals, and the coefficient of variation. Significance was set at $p \leq 0.05$.

Results

All players completed the measurements without injuries. Table 1 presents the results from the reliability analysis for all measurements. In addition, Table 2 presents the comparison between SG and WG. No significant difference was found for age, body mass, body height, BMI, and FFM between groups. However, SG had 23.5% lower-body fat compared with WG. Moreover, SG had 7.4 and 8.2% faster throwing velocity from 7 and 9 m line, respectively, compared with WG. The stronger group had −6.3% and −7.3% lower time-trials in 0–20 m sprint and T-half test, respectively, compared with WG. The stronger group had 10.9%

Table 1
Intraclass correlation coefficients for all variables.*

Variables	ICC	Lower bound	Upper bound	p
Body mass	0.998	0.998	0.999	0.001
Percent body fat	0.990	0.982	0.995	0.001
Fat-free mass	0.965	0.958	0.987	0.005
CMJ height	0.989	0.957	0.997	0.001
IPT	0.990	0.964	0.998	0.001
RTD	0.893	0.649	0.972	0.01
5-step long jump	0.985	0.970	0.997	0.005
0–20 m sprint	0.950	0.860	0.980	0.001
T-half test	0.940	0.820	0.96	0.001
Throwing velocity	0.970	0.900	0.990	0.001
1RM in bench press	0.966	0.914	0.985	0.001
1RM in squat	0.960	0.810	0.940	0.001

*ICC = intraclass correlation coefficient; CMJ = countermovement jump; IPT = isometric peak torque; RTD = rate of torque development; 1RM = 1 repetition maximum.

higher 5 step long-jump performance and 21.5% CMJ height compared with their weaker counterparts. Finally, SG had 22.4% greater IPT and 20.5 and 36.2% greater 1RM strength in bench press and squat, respectively, compared with WG.

Significant differences were found for the isometric leg extension measurement. Figure 1 presents the torque-time curves for absolute and relative to body mass values. The stronger group had higher torque at 120 milliseconds (13.0%, $p = 0.045$, $g = 0.856$), 150 milliseconds (14.8%, $p = 0.031$, $g = 0.931$), 200 milliseconds (16.8%, $p = 0.024$, $g = 0.975$), and 250 milliseconds (19.5%, $p = 0.015$, $g = 1.062$), compared with WG (Figure 1A). In addition, when torque was expressed relative to body mass, SG had significant differences at 100 milliseconds (12.0%, $p = 0.031$, $g = 0.932$), 120 milliseconds (13.1%, $p = 0.022$, $g = 1.004$), 150 milliseconds (15.1%, $p = 0.011$, $g = 1.128$), 200 milliseconds (17.2%, $p = 0.002$, $g = 1.423$), and 250 milliseconds (20.3%, $p = 0.001$, $g = 1.624$), compared with WG (Figure 1B). Similarly, Figure 2 presents the differences between SG and WG for RTD. The Stronger group had greater RTD in absolute values at 120, 150, 200, and 250 milliseconds compared with WG (Figure 2A). Moreover, when RTD was expressed relative to body mass, SG had greater performance at 100, 120, 150, 200, and 250 milliseconds, compared with WG (Figure 2B).

Table 3 presents the correlation coefficients between throwing velocity, sprinting, change of direction, CMJ, IPT, and 1RM strength. Performance in throwing velocity both from 7 and 9 m line was significantly correlated with IPT and 1RM strength. In addition, sprinting performance and change of direction were significantly correlated with CMJ height and 5-step long jump. Five-step long jump was significantly correlated with 1RM squat and CMJ height.

Table 4 presents the correlations between body composition and performance variables. Fat-free mass was significantly correlated with throwing velocity (Figure 3), IPT, and 1RM strength in bench press and squat. Percentage body fat had a negative impact on sprinting, change of direction, and CMJ.

Table 5 presents the correlation coefficients between RTD and performance in throwing velocity, sprinting, CMJ, 1RM strength, and body composition. Throwing velocity from 7 and 9 m line was significantly correlated with RTD from 60 milliseconds until 250 milliseconds (Figure 4). The rate of torque development also correlated with IPT, 1RM strength, and fat-free mass.

Multiple regression analysis revealed that throwing velocity from the 7 m line could be explained by the linear combination of fat-free mass (beta coefficient: 0.727, $p = 0.001$) and T-half test change of direction time trial (beta coefficient: -0.464, $p = 0.004$) by $R^2_{adj} = 0.601$ ($p = 0.001$). Similarly, throwing velocity from the 9 m line could be explained by the linear combination of fat-free mass (beta coefficient: 0.735, $p = 0.001$), T-half test change of direction time trial (beta coefficient: -0.869, $p = 0.001$), and 5-step long jump (beta coefficient: -0.510, $p = 0.031$) by $R^2_{adj} = 0.657$ ($p = 0.001$).

Discussion

The purpose of the study was to investigate the influence of strength level in handball-specific fitness elements between stronger and weaker athletes as well as to examine the relationship between strength and RTD with handball-specific fitness elements. The main finding of the study was that players with higher strength level had significantly higher performance on handball-specific fitness elements compared with weaker players. More specific, SG had significantly higher performance in CMJ height, IPT, and RTD both in absolute and relative to body mass compared with WG, whereas SG had lower 0–20 linear sprint and change of direction time trials, lower CoDd, but higher 5-step horizontal jump and throwing velocity compared with WG. Consequently, stronger players may have a greater advantage in handball-specific fitness elements compared with their weaker counterparts and potentially a greater on court performance during competition. In addition, 1RM strength and IPT were largely correlated with jumping performance and throwing velocity, but the correlations with sprinting and change of direction were trivial to moderate. One interesting finding of the study was that RTD contributed significantly to throwing velocity, leading to the conclusion that fast force production, as measured in the current study with the isometric leg extension test, is also a major parameter that may influence performance in handball. These results suggest that coaches and strength and conditioning professionals may design training programs aiming at

Table 2
Differences between stronger ($N = 10$) and weaker ($N = 11$) players.*

Variable	SG	WG	p	Hedges' g
Age (y)	27.0 ± 5.98	24.8 ± 7.9	0.489	0.280
Body mass (kg)	87.5 ± 11.8	88.4 ± 16.1	0.891	0.056
Body height (m)	1.83 ± 0.09	1.80 ± 0.07	0.449	0.306
BMI (kg·m^{-2})	26.1 ± 2.70	26.5 ± 4.93	0.835	0.085
FFM (kg)	74.1 ± 7.4	70.5 ± 8.9	0.325	0.406
Fat (%)	14.9 ± 4.8	19.4 ± 5.4	0.054	0.822
7 m throw (m·s^{-1})	23.33 ± 1.5	21.7 ± 1.1	0.009	1.161
9 m throw (m·s^{-1})	25.0 ± 1.6	23.1 ± 1.3	0.008	1.185
0–20 sprint (s)	3.17 ± 0.16	3.38 ± 0.19	0.013	1.100
T-half test (s)	5.98 ± 0.28	6.45 ± 0.28	0.001	1.527
CoDd (s)	2.81 ± 0.26	3.07 ± 0.16	0.011	1.106
5-step long jump (m)	13.10 ± 1.05	11.82 ± 0.68	0.003	1.323
CMJ height (cm)	42.09 ± 4.38	34.65 ± 4.83	0.002	1.474
Bench press 1RM (kg)	94.4 ± 13.9	78.4 ± 18.9	0.040	0.885
Squat 1RM (kg)	157.1 ± 19.0	115.4 ± 26.4	0.001	1.658
IPT (N·m^{-1})	407.6 ± 61.8	333.1 ± 52.5	0.008	1.187

*SG = strong group; WG = weak group; BMI = body mass index; FFM = fat-free mass; CODd = change of direction deficit; CMJ = countermovement jump; IPT = isometric peak torque.

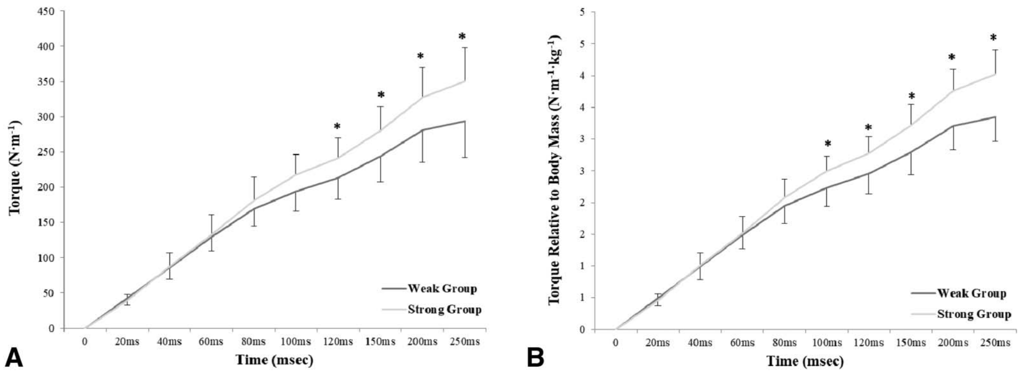

Figure 1. (A) Leg extension torque in absolute values. Stronger players had greater torque at 120, 150, 200, and 250 milliseconds compared with weaker players. (B) Leg extension torque relative to body mass. Stronger players had greater torque at 100, 120, 150, 200, and 250 milliseconds compared with weaker players. *Significant difference between stronger and weaker players.

increasing muscle strength and fast force production in handball players.

Stronger players had greater performance in all handball-specific fitness tests compared with weaker players, whereas no significant difference was found for body composition, although there was a trend for statistical significance between groups for body fat percentage ($p = 0.054$). According to the best of the author's knowledge, this is the first study to compare stronger and weaker handball players in sport-specific fitness elements. A previous study had showed that there is a significant difference in physical fitness parameters and body composition between elite and amateur handball players. More specific, a study in male handball players showed that elite players had greater 1RM strength in bench press, higher muscle power in bench press and squat, higher throwing velocity, 3-step running velocity, and higher values in body mass and fat-free mass compared with amateur players, although no significant differences were observed for sprint, vertical jump, and body fat (11). These differences in strength, CMJ height, and throwing velocity were partly explained by the significant differences in body mass and fat-free mass between elite and amateur players, a result which was not confirmed in this study. On the contrary, a study which evaluated 44 young male handball players showed that elite players had higher sprint and vertical jump performance, strength, and throwing velocity compared with players younger than 18 and 16 (29). In this study, no significant difference between SG and WG was found for body composition. However, stronger players had the ability to move faster, change direction rapidly, and achieve greater throwing velocities compared with weaker players. Consequently, the level of muscle strength may also be a noteworthy factor justifying the separation between higher and lower performance players. One interesting finding of this study was the significant difference in CoDd between SG and WG, which had been previously proposed as the additional time that one directional change requires when compared with a pure linear sprint over an equivalent distance (28). Stronger players not only changed direction rapidly, but they had also lower CoDd time compared with weaker players leading to the conclusion that stronger players may have a greater change of direction ability regardless of the linear sprint performance compared with weaker players. It has also been suggested that performance in CoDd may be considered as an independent variable (27); thus, handball coaches may also use the CoDd to evaluate players change of direction ability combined with linear and change of direction performance tests. This is the first study to evaluate the role of CoDd on handball players; thus, more research is needed to investigate the role of CoDd as a handball sport-specific test.

Stronger players had greater RTD and IPT compared with their weaker counterparts. Both neural and muscle factors may affect RTD, whereas maximum strength and fat-free mass are vital components that significantly contribute to fast force/

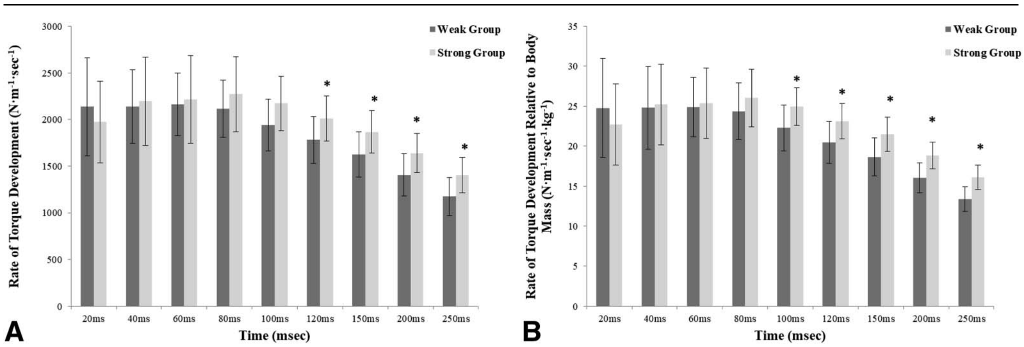

Figure 2. (A) Leg extension isometric rate of torque development (RTD) in absolute values. Stronger players had a greater rate of torque development at 120, 150, 200, and 250 milliseconds compared with weaker players. (B) Leg extension RTD relative to body mass. Stronger players had greater RTD at 100, 120, 150, 200, and 250 milliseconds compared with weaker players. *Significant difference between stronger and weaker players.

Table 3

Correlations between throwing velocity, sprinting, change of direction, jumping, isometric peak force, and 1RM strength in 21 handball players.*

	9 m throw	T-half test	0–20 m sprint	CoDd	CMJ height	5-step long jump	IPT	1RM bench press	1RM squat
7 m throw	0.871‡	−0.353	−0.289	−0.286	0.256	0.200	0.819‡	0.472†	0.639‡
9 m throw	1	−0.351	−0.159	−0.391	0.270	0.128	0.772‡	0.512†	0.580‡
T-half test		1	0.773‡	0.849‡	−0.789‡	−0.796‡	−0.281	−0.030	−0.399
0–20 m sprint			1	0.321	−0.751‡	−0.699‡	−0.159	0.038	−0.414
CoDd				1	−0.562‡	−0.607‡	−0.286	−0.076	−0.250
CMJ height					1	0.783‡	0.231	0.183	0.519†
5-step long jump						1	0.227	0.013	0.524†
IPT							1	0.595‡	0.762‡
1RM bench press								1	0.661‡

*CMJ = countermovement jump; IPT = isometric peak torque; CoDd = change of direction deficit; 1RM = 1 repetition maximum.
†$p < 0.05$.
‡$p < 0.01$.

torque production (6,22). Although no significant differences were found for fat-free mass between SG and WG, the differences in maximum strength and IPT combined with the possible differences in the neuromuscular level between groups may explain the differences in RTD. A previous study in male collegiate athletes (cricket, judo, rugby, and soccer) showed that stronger athletes had greater maximum RFD and peak force relative to body mass compared with weaker athletes in isometric midthigh pull testing (34). Similar results were found between stronger and weaker netball players for lower-body isometric midthigh pull force (35). In addition, a recent study in male individuals showed that stronger subjects had greater upper-body RFD and IPF as evaluated from isometric bench press, compared with weaker subjects (42), meaning that the level of strength may have a significant influence on both lower-body and upper-body RFD. Considering that increases in muscle strength may lead to increases in RFD, then coaches should regularly perform strength training with maximum intentional movement velocity (1) in an attempt to enhance RFD and as a consequence on court handball performance. Still, more research is required to investigate the longitudinal effects of training on RFD and the possible changes after different training programs in handball players.

Correlational analysis revealed significant connections between strength and handball-specific fitness elements. Maximum strength in squat was largely correlated with CMJ height, whereas the correlation with IPT was very large. On the contrary, a study in 12 elite male Spanish handball players found no significant correlation between isokinetic peak torques (60–180°) and jumping variables (9). The discrepancy between the 2 studies may be due to the level of the athletes and to the nature of measurement (isometric vs. isokinetic evaluation). In addition, moderate to very large correlations were found between 1RM strength in bench press, squat, and IPT with throwing velocity. Similarly, a study in male Tunisian national handball league players showed that 1RM in bench press and lower-body force production were significantly correlated with throwing velocity (3), whereas significant correlations were also found between squat and throwing velocity in another study on senior Tunisian national handball players (2). Moreover, a study in elite male Spanish players showed significant correlation between 3-step throwing velocity with different percentage velocity of 1RM bench press and half squat (11). Although strength is a significant factor that contributes to throwing velocity and jumping performance in handball players, trivial-to-moderate correlations were found between 1RM strength and sprinting and change of direction, results that are similar to a previous study (2). This was further reinforced by the small correlations between fat-free mass with sprinting and change of direction. Sprinting and change of direction involve the movement of almost all muscle groups with maximum speed; thus, strength may not be a good predictor for fast on court movements. However, the linear combination of the T-half test and fat-free mass could significantly explain the 60% of throwing velocity from 7 m, whereas the linear combination of T-half test, fat-free mass, and 5-step horizontal jump could significantly explain the 66% of throwing velocity from 9 m. The ability to change directions rapidly is an index of a well-trained muscular system and shows that the particular player has the ability to apply his force rapidly on the ground. Consequently, regardless of the movement during the throw (standing or jumping shot), the transition of ground forces to the upper-body depends on lower-body fast force production. This, combined with a high fat-free mass may lead to a faster ball throwing velocity. Thus, change of direction, long jump, and fat-free mass may be good

Table 4

Correlations between body composition and throwing velocity, jumping, sprinting, change of direction, IPT, and 1RM strength in 21 handball players.*

	7 m throw	9 m throw	T-half test	0–20 m sprint	CoDd	CMJ height	5-step long jump	IPT	1RM bench press	1RM squat
BMI	0.123	0.060	0.519†	0.508†	0.351	−0.448†	−0.236	0.405	0.292	0.200
FFM	0.656‡	0.639‡	0.153	0.256	0.045	−0.194	−0.073	0.774‡	0.518†	0.544†
Fat	0.235	0.041	0.576‡	0.509†	0.436†	−0.603‡	−0.429	0.286	0.001	−0.017

*CoDd = change of direction deficit; CMJ = countermovement jump; IPT = isometric peak torque; BMI = body mass index; FFM = fat-free mass.
†$p < 0.05$.
‡$p < 0.01$.

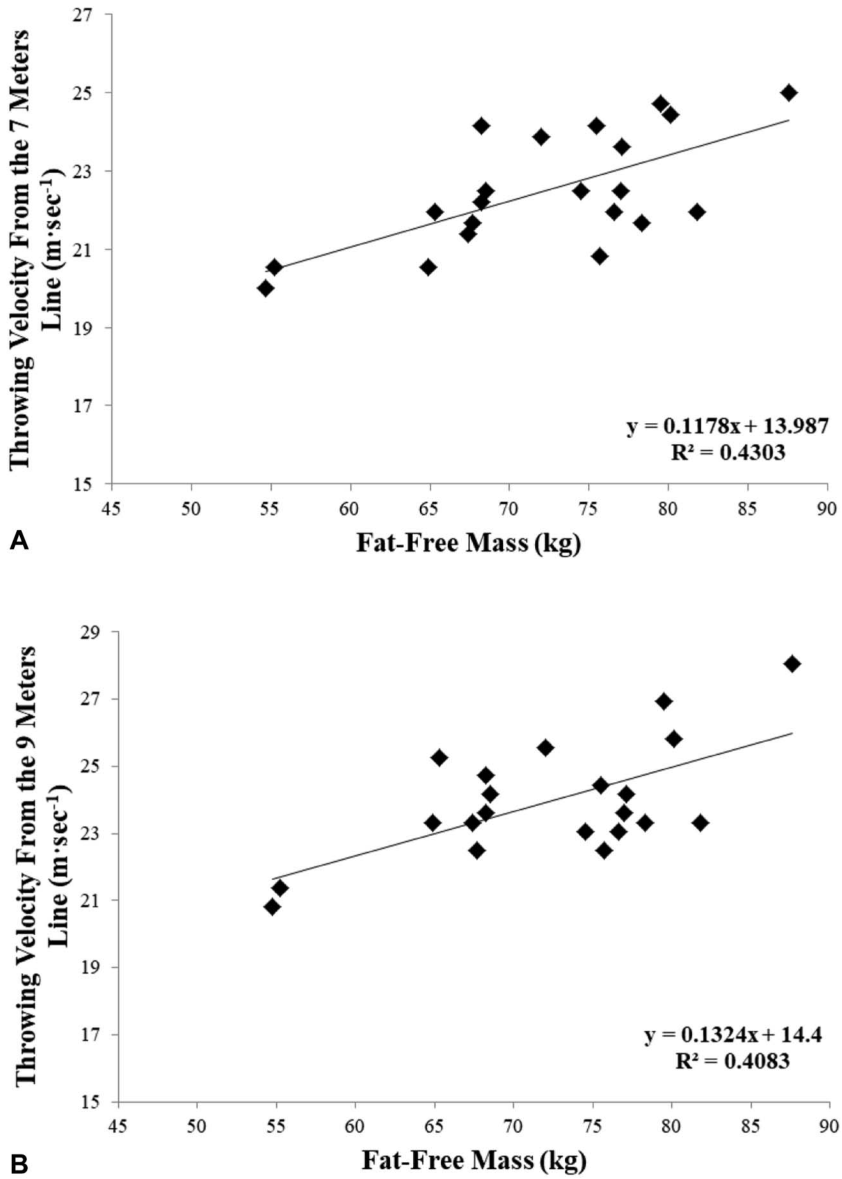

Figure 3. Correlations between fat-free mass with standing throwing velocity from 7 m (A) and jumping shot throwing velocity from the 9 m line (B).

predictors for throwing velocity; hence, coaches should focus on increasing these factors to enhance throwing velocity.

A novel finding of the current study was that RTD was significantly correlated with throwing velocity, 1RM strength, and fat-free mass. Several studies have focus on the correlation between upper-body and lower-body isokinetic torque with throwing velocity and jumping performance and found trivial correlations (9,38). In addition, results from a study in elite male handball players showed that upper-limb RFD calculated during a concentric-only bench press exercise was not correlated with throwing velocity (25). These findings are in disagreement with this study as very large correlations were found between throwing velocity and RTD especially during the late phase of the torque-time curve (>100 milliseconds). The late phase of RTD is mainly determined by the amount of lean mass, the fiber type composition, and the maximum strength (22,26). Interestingly, fat-free mass largely correlated with RTD, maximum strength, and throwing velocity. Consequently, increasing fat-free mass by resistance training, coaches may enhance RTD and fast torque production leading to a higher on court performance. However, according to the best of the author's knowledge, this is the first study which investigated the possible correlation between RTD with handball-specific fitness elements; hence, more studies are needed to investigate whether upper-body or lower-body RTD may be a strong predictor for handball-specific fitness elements.

In the current study, all measurements were performed during the end of the preseason period; thus, players might not be in their best physical/competitive level. In addition, electromyography signals and other biological factors that could provide better insights into the nature of the current results derived were not possible to be evaluated. The small sample size especially for the comparison between stronger and weaker players may limit the generalization of the results. Only male players participated in the study; thus, other studies

Table 5

Correlation coefficients between the rate of torque development in different time windows and throwing velocity, sprinting, change of direction, jumping, and 1RM strength in 21 handball players.*

	RTD 20 ms	RTD 40 ms	RTD 60 ms	RTD 80 ms	RTD 100 ms	RTD 120 ms	RTD 150 ms	RTD 200 ms	RTD 250 ms
7 m throw	0.069	0.351	0.525†	0.599‡	0.753‡	0.776‡	0.792‡	0.837‡	0.835‡
9 m throw	−0.063	0.192	0.336	0.418	0.579‡	0.596‡	0.647‡	0.747‡	0.780‡
T-half test	0.080	−0.241	−0.130	−0.224	−0.249	−0.305	−0.361	−0.285	−0.291
0–20 m sprint	0.144	−0.138	−0.071	−0.120	−0.142	−0.169	−0.215	−0.125	−0.115
CoDd	0.002	−0.245	−0.135	−0.235	−0.253	−0.315	−0.360	−0.321	−0.339
CMJ height	−0.077	0.203	0.111	0.193	0.238	0.257	0.282	0.231	0.250
5-step long jump	0.207	0.429	0.280	0.378	0.383	0.402	0.428	0.310	0.280
IPT	0.059	0.381	0.592‡	0.695‡	0.824‡	0.815‡	0.823‡	0.893‡	0.915‡
1RM bench press	0.098	0.223	0.327	0.397	0.486†	0.459†	0.470†	0.549‡	0.553‡
1RM squat	0.013	0.402	0.494†	0.671‡	0.787‡	0.788‡	0.794‡	0.768‡	0.749‡
BMI	0.276	0.147	0.289	0.270	0.314	0.303	0.286	0.353	0.377
FFM	0.218	0.346	0.540†	0.609‡	0.715‡	0.694‡	0.685‡	0.753‡	0.760‡
Fat	0.247	0.179	0.394	0.339	0.351	0.330	0.298	0.322	0.300

*RTD = rate of torque development; CMJ = countermovement jump; CoDd = change of direction deficit; IPT = isometric peak torque; BMI = body mass index; FFM = fat-free mass.
†$p < 0.05$.
‡$p < 0.01$.

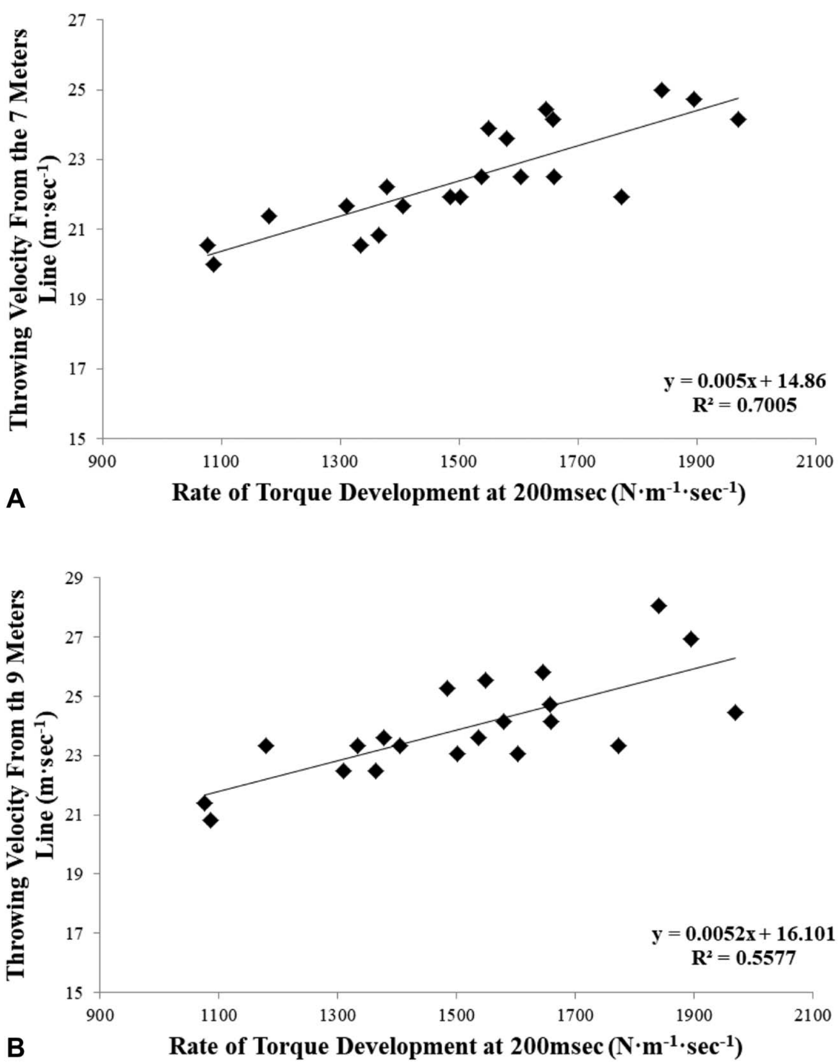

Figure 4. Correlations between leg extension isometric rate of torque development at 200 milliseconds with standing throwing velocity from 7 m (A) and jumping shot throwing velocity 9 m line (B).

should focus on female players as well. However, all players participated in the study were at the highest level of competition in the national league. Future studies should focus on the long-term application of strength training and the development of strength levels to further observe the relationship of strength levels with speed, agility, and other handball-specific fitness elements and to identify the highest point of this relationship.

In conclusion, stronger players may have a greater performance in handball-specific fitness elements compared with their weaker counterparts. The study showed that the level of muscle strength may determine the performance in all fitness attributes. In addition, 1RM strength was largely correlated with throwing velocity and jumping performance, but the correlations with sprinting and change of direction were trivial to moderate, in professional handball players. Moreover, RTD and fat-free mass may be significant determinants of throwing velocity and on court performance in handball.

Practical Applications

Handball coaches and strength and conditioning practitioners should aim at the acquisition of high strength levels within their handball teams for achieving a superior on court performance. With this as a goal, strength training should be a vital part of training for all players but mainly for weaker players, who are suggested to spend more training time on resistance training programs aiming in increasing their strength levels and fat-free mass. In addition, resistance training should be performed with maximum intentional movement velocity especially during the concentric phase of movement to enhance RFD which, as was demonstrated in this study, was significantly correlated with throwing velocity. At the same time, as increases in muscle strength occur, coaches should regularly monitor changes in fat-free mass which seems to be a significant contributing factor not only for strength and RFD but for throwing velocity as well.

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References

1. Blazevich AJ, Wilson CJ, Alcaraz PE, Rubio-Arias JA. Effects of resistance training movement pattern and velocity on isometric muscular rate of force development: A systematic review with meta-analysis and meta-regression. *Sports Med* 50: 943–963, 2020.
2. Chaouachi A, Brughelli M, Levin G, et al. Anthropometric, physiological and performance characteristics of elite team-handball players. *J Sports Sci* 27: 151–157, 2009.
3. Chelly MS, Hermassi S, Shephard RJ. Relationships between power and strength of the upper and lower limb muscles and throwing velocity in male handball players. *J Strength Cond Res* 24: 1480–1487, 2010.
4. Chelly MS, Hermassi S, Aouadi R, et al. Match analysis of elite adolescent team handball players. *J Strength Cond Res* 25: 2410–2417, 2011.
5. Cormier P, Freitas TT, Rubio-Arias JÁ, Alcaraz PE. Complex and contrast training: Does strength and power training sequence affect performance-based adaptations in team sports? A systematic review and meta-analysis. *J Strength Cond Res* 34: 1461–1479, 2020.
6. Del Vecchio A, Negro F, Holobar A, et al. You are as fast as your motor neurons: Speed of recruitment and maximal discharge of motor neurons determine the maximal rate of force development in humans. *J Physiol* 597: 2445–2456, 2019.
7. Elias LJ, Bryden MP, Bulman-Fleming MB. Footedness is a better predictor than is handedness of emotional lateralization. *Neuropsychologia* 36: 37–43, 1998.
8. Fritz C, Morris P, Richler J. Effect size estimates: Current use, calculations, and interpretation. *J Exp Psychol Gen* 141: 2–18, 2012.
9. González-Ravé JM, Juárez D, Rubio-Arias JA, et al. Isokinetic leg strength and power in elite handball players. *J Hum Kinet* 41: 227–233, 2014.
10. Gorostiaga EM, Izquierdo M, Iturralde P, Ruesta M, Ibáñez J. Effects of heavy resistance training on maximal and explosive force production, endurance and serum hormones in adolescent handball players. *Eur J Appl Physiol Occup Physiol* 80: 485–493, 1999.
11. Gorostiaga EM, Granados C, Ibáñez J, Izquierdo M. Differences in physical fitness and throwing velocity among elite and amateur male handball players. *Int J Sports Med* 91: 698–707, 2005.
12. Gorostiaga EM, Granados C, Ibáñez J, González-Badillo JJ, Izquierdo M. Effects of an entire season on physical fitness changes in elite male handball players. *Med Sci Sports Exerc* 38: 357–366, 2006.
13. Granados C, Izquierdo M, Ibáñez J, Bonnabau H, Gorostiaga EM. Differences in physical fitness and throwing velocity among elite and amateur female handball players. *Int J Sports Med* 28: 860–867, 2007.
14. Hopkins WG. Measures of reliability in sports medicine and science. *Sports Med* 30: 1–15, 2000.
15. Hermassi S, Chelly MS, Tabka Z, Shephard RJ, Chamari K. Effects of 8-week in-season upper and lower limb heavy resistance training on the peak power, throwing velocity and sprint performance of elite male handball players. *J Strength Cond Res* 25: 2424–2433, 2011.
16. Hermassi S, Ingebrigtsen J, Schwesig R, et al. Effects of in-season short-term aerobic and high intensity interval training program on repeated sprint ability and jump performance in handball players. *J Sports Med Phys Fitness* 58: 50–56, 2018.
17. Hermassi S, Schwesig R, Aloui G, Shephard RJ, Chelly MS. Effects of short-term in-season weightlifting training on the muscle strength, peak power, sprint performance, and ball-throwing velocity of male handball players. *J Strength Cond Res* 33: 3309–3321, 2019.
18. Jaggers JR, Swank AM, Frost KL, Lee CD. The acute effects of dynamic and ballistic stretching on vertical jump height, force, and power. *J Strength Cond Res* 22: 1844–1849, 2008.
19. Karcher C, Buchheit M. On-court demands of elite handball, with special reference to playing positions. *Sports Med* 44: 797–814, 2014.
20. Loturco I, Nimphius S, Kobal R, et al. Change-of direction deficit in elite young soccer players. *German J Exerc Sport Res* 48: 228–234, 2018.
21. Maden-Wilkinson TM, Balshaw TG, Massey GJ, Folland JP. What makes long-term resistance-trained individuals so strong? A comparison of skeletal muscle morphology, architecture, and joint mechanics. *J Appl Physiol* 128: 1000–1011, 2020.
22. Maffiuletti NA, Aagaard P, Blazevich AJ, et al. Rate of force development: Physiological and methodological considerations. *Eur J Appl Physiol* 116: 1091–1116, 2016.
23. Marques MC, van den Tilaar R, Vescovi JD, Gonzalez-Badillo JJ. Relationship between throwing velocity, muscle power, and bar velocity during bench press in elite handball players. *Int J Sports Physiol Perform* 2: 414–422, 2007.
24. Marques MC. In-season strength and power training for professional male team handball players. *Strength Cond J* 32: 74–81, 2010.
25. Marques MC, Saavedra F, Abrantes C, Aidar F. Associations between rate of force development metrics and throwing velocity in elite team handball players: A short research report. *J Hum Kinet* 29A: 53–57, 2011.
26. Methenitis S, Spengos K, Zaras N, et al. Fiber type composition and rate of force development in endurance- and resistance-trained individuals. *J Strength Cond Res* 33: 2388–2397, 2019.
27. Nimphius S, Geib G, Spiteri T, Carlisle D. Change of direction deficit measurement in Division I American football players. *J Aust Strength Cond* 21: 115–117, 2013.
28. Nimphius S, Callaghan SJ, Spiteri T, Lockie RG. Change of direction deficit: A more isolated measure of change of direction performance than total 505 time. *J Strength Cond Res* 30: 3024–3032, 2016.
29. Ortega-Becerra M, Pareja-Blanco F, Jiménez-Reyes P, Cuadrado-Penãfiel V, González-Badillo JJ. Determinant factors of physical performance and specific throwing in handball players of different ages. *J Strength Cond Res* 32: 1778–1786, 2018.
30. Panteli N, Hadjicharalambous M, Zaras N. Delayed potentiation effect on sprint, power and agility performance in well-trained soccer players. *J Sci Sport Exerc* Online ahead of print 2023.
31. Sassi RH, Dardouri W, Yahmed MH, et al. Relative and absolute reliability of a modified agility t-test and its relationship with vertical jump and straight sprint. *J Strength Cond Res* 23: 1644–1651, 2009.

32. Secomb JL, Nimphius S, Farley O-R, et al. Lower-body muscle structure and jump performance of stronger and weaker surfing athletes. *Int J Sports Physiol Perform* 11: 652–657, 2016.
33. Turner AN, Comfort P, McMahon J, et al. Developing powerful athletes, part 1: Mechanical underpinnings. *Strength Cond J* 42: 30–39, 2020.
34. Thomas C, Jones PA, Rothwell J, Chiang CY, Comfort P. An investigation into the relationship between maximum isometric strength and vertical jump performance. *J Strength Cond Res* 29: 2176–2185, 2015.
35. Thomas C, Comfort P, Jones PA, Dos Santos T. A comparison of isometric midthigh-pull strength, vertical jump, sprint speed, and change-of-direction speed in academy netball players. *Int J Sports Physiol Perform* 12: 916–921, 2017.
36. Torres-Luque G, Calahorro-Cañada F, Lara-Sánchez AJ, Garatachea N, Nikolaidis PT. Body composition using bioelectrical impedance analysis in elite young soccer players: The effects of age and playing position. *Sport Sci Health* 11: 203–210, 2015.
37. Wagner H, Finkenzeller T, Würth S, Von Duvillard SP. Individual and team performance in team-handball: A review. *J Sports Sci Med* 13: 808–816, 2014.
38. Zapartidis I, Gouvali M, Bayios I, Boudolos K. Throwing effectiveness and rotational strength of the shoulder in team handball. *J Sports Med Phys Fitness* 47: 169–178, 2007.
39. Zapartidis I, Skoufas D, Vareltzis I, et al. Factors influencing ball throwing velocity in young female handball players. *Open Sports Med J* 3: 39–43, 2009.
40. Zaras ND, Stasinaki A-N, Methenitis SK, et al. Rate of force development, muscle architecture and performance in young competitive track and field throwers. *J Strength Cond Res* 30: 81–92, 2016.
41. Zaras N, Stasinaki A-N, Spiliopoulou P, et al. Rate of force development, muscle architecture, and performance in elite weightlifters. *Int J Sports Physiol Perform* 16: 216–223, 2021.
42. Zaras N, Stasinaki A-N, Spiliopoulou P, Mpampoulis T, Terzis G. Triceps brachii muscle architecture, upper-body rate of force development, and bench press maximum strength of strong and weak trained participants. *Hum Mov* 24: 121–129, 2023.
43. Ziv G, Lidor R. Physical characteristics, physiological attributes, and on court performances of handball players: A review. *Eur J Sport Sci* 9: 375–386, 2009.